WORKED SOLUTION

STEP I 1992, Question 4

Pure • Loci in the Argand Diagram

STEP I 1992, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q4

The key idea

Turn each condition into a statement about x and y (with $z = x + \mathrm{i}y$), or about the modulus and argument of z, and the region usually becomes obvious. Parts (i) and (ii) need nothing more than this. Part (iii) is different: it describes a set of *sums*, so we have to show two things. Every sum lies in the region we claim, and every point of that region really is such a sum.

Solution

Throughout, $z = x + \mathrm{i}y$ with x, y real.

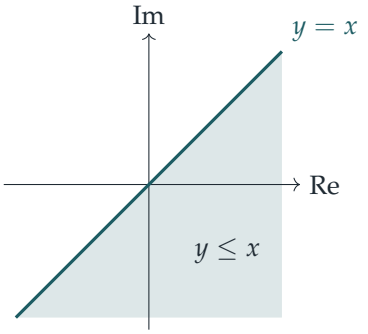
Part (i)

Expanding,

$$(1+\mathrm{i})z = (1+\mathrm{i})(x+\mathrm{i}y) = (x-y) + \mathrm{i}(x+y),$$

so $\mathrm{Re}((1+\mathrm{i})z) = x - y$. The condition is $x - y \geq 0$, that is $y \leq x$. The set is the closed half-plane on and below the line $y = x$, including the line itself.

Geometrically, multiplying by $1 + \mathrm{i} = \sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$ rotates by $\frac{\pi}{4}$ and scales by $\sqrt{2}$. So the condition says that z, rotated anticlockwise through $\frac{\pi}{4}$, lands in the right half-plane $\mathrm{Re} \geq 0$. That means $-\frac{3\pi}{4} \leq \arg z \leq \frac{\pi}{4}$ (or $z = 0$), which is the same half-plane.



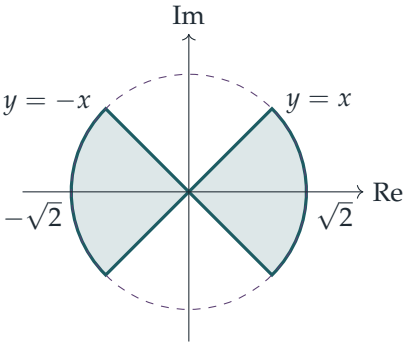
Part (ii)

Since $|z^2| = |z|^2$, the first condition is $|z|^2 \leq 2$, that is $|z| \leq \sqrt{2}$: the closed disc of radius $\sqrt{2}$ centred at the origin.

For the second, $z^2 = (x^2 - y^2) + 2ixy$, so $\mathrm{Re}(z^2) = x^2 - y^2$ and the condition is $x^2 \geq y^2$, that is $|y| \leq |x|$. This is the pair of opposite closed sectors between the lines $y = x$ and $y = -x$ that contain the real axis.

In polar form the same thing is quicker: if $z = re^{i\theta}$ then $z^2 = r^2 e^{2i\theta}$, so we need $r \leq \sqrt{2}$ and $\cos 2\theta \geq 0$, which gives $-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}$ or $\frac{3\pi}{4} \leq \theta \leq \frac{5\pi}{4}$ (or $z = 0$).

The set is the intersection: two closed quarter-discs (each a sector of angle $\frac{\pi}{2}$ and radius $\sqrt{2}$), one opening to the right and one to the left, meeting at the origin. The boundary is included throughout.



Part (iii)

Let S be the set of sums. We claim S is the closed annulus $\{z : 1 \leq |z| \leq 3\}$.

Every sum lies in the annulus. If $|z_1| = 2$ and $|z_2| = 1$, the triangle inequality gives

$$|z_1 + z_2| \leq |z_1| + |z_2| = 3, \qquad |z_1 + z_2| \geq |z_1| - |z_2| = 1.$$

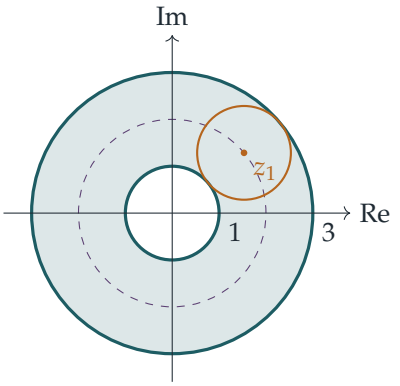
Ask yourself Is that enough? How do we know the whole annulus is filled, and not just some curve inside it?

The inequalities only show that S lies *inside* the annulus, so we still need the reverse inclusion. Take any z with $1 \le |z| \le 3$. We want a unit complex number z_2 with $|z - z_2| = 2$, because then $z_1 = z - z_2$ has modulus 2 and $z = z_1 + z_2$.

Geometrically, z_2 must lie on the unit circle centred at 0 and also on the circle of radius 2 centred at z. Two circles with radii 1 and 2 whose centres are a distance $d = |z|$ apart meet exactly when $2 - 1 \le d \le 2 + 1$, and that is our condition $1 \le |z| \le 3$.

To see this without quoting the circle fact, write $z = re^{i\alpha}$ and consider $f(\varphi) = |z - e^{i\varphi}|$, which is continuous in φ. At $\varphi = \alpha$ the point $e^{i\varphi}$ points the same way as z, so $f(\alpha) = |r - 1| = r - 1 \le 2$. At $\varphi = \alpha + \pi$ it points the opposite way, so $f(\alpha + \pi) = r + 1 \ge 2$. By the intermediate value theorem $f(\varphi) = 2$ for some φ, and $z_2 = e^{i\varphi}$ works.

So S is exactly the closed annulus between the circles $|z| = 1$ and $|z| = 3$, with both boundary circles included. A good way to picture it: fix z_1 on the circle of radius 2, and $z_1 + z_2$ traces a unit circle centred at z_1. As z_1 goes round, these little circles sweep out the annulus.



The amber circle shows the sums for one fixed z_1 on the dashed circle $|z_1| = 2$.

Checks

- In (i), $z = 1$ gives $\mathrm{Re}(1 + i) = 1 \ge 0$, and $z = i$ gives $\mathrm{Re}(-1 + i) = -1 < 0$. This agrees with $y \le x$.
- In (ii), $z = 1$ is in the set and $z = i$ is not ($i^2 = -1$), so the sectors should contain the real axis, as drawn.
- In (iii), $z_1 = 2$, $z_2 = -1$ gives 1 and $z_1 = 2$, $z_2 = 1$ gives 3, so both boundary circles are reached. We tested all three descriptions on 200 000 random points. Random sums had moduli between 1.0000 and 3.0000, and for 1000 random points of the annulus a suitable z_2 was found to within 2×10^{-4}.

Common slips

- In (ii), keeping only the sector $|\arg z| \leq \frac{\pi}{4}$ and forgetting the left-hand one. Both z and $-z$ have the same square.
- In (iii), stopping after the triangle inequality. That shows only that S lies inside the annulus, not that S is the whole annulus.
- Leaving out whether boundaries are included. Every inequality here is non-strict, so every boundary is in.

Beyond STEP

Part (iii) is a small instance of the *Minkowski sum* $A + B = \{a + b : a \in A,\ b \in B\}$ of two sets, an idea that runs through geometry and convex analysis. The two-inclusion argument (show $S \subseteq T$, then $T \subseteq S$) is the standard way to prove that two sets are equal, and you will use it constantly at university. For a geometric view of complex multiplication as rotation and scaling, which makes (i) and (ii) almost instant, see Needham, *Visual Complex Analysis*.

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